Data Science

Relevel
by Unacademy

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About Relevel by Unacademy

The coming decade is one of immense opportunity for India with a majority of our population under the age of 35 and the growing proliferation of global technology.

For a long time, India was known as the land of outsourced opportunities. The past decade however has shown that Indians can build global products and businesses. This is expected to grow manifold in the coming years provided we solve for one crucial element - skilled talent.

Today, there is a massive divide between aspirants looking for the right opportunities on one side, and companies vying for the best talent on the other. Historically exposure to opportunities has been defined by one's college, degree, or who they already knew in the industry. The events over the last year have only made the situation worse.

We believe it is high time that this changes for good! We believe that hiring should depend only on your capabilities and skills, not your college degree.

Being India's Largest Learning Platform, democratizing knowledge and access is a core tenet of the Unacademy Group. We are now extending that to opportunities through Relevel by leveling the playing field for millions of Indians. We imagine a future where candidates have a transparent path to the career of their dreams.

At Relevel, we believe your history doesn't matter, your skills do. Your network doesn't matter, you do!

All the best. We know you can crack it!

Why become a Data Scientist?

As a Data Scientist, you have the opportunity to support your organization's success through data-driven insights. It's a career where every day brings new challenges and new ways to put your skills into practice.

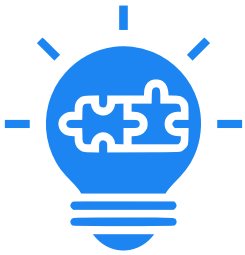
Value to business: Each and every bit of data is crucial to an organization and so is its analysis. Data scientists are needed across industries from healthcare to e-commerce. Data scientists are the go-to professionals for management teams as they help answer very strategic questions using their analytical and technical skills.

Enormous demand: Data Scientist is a comparatively new profile but one which is in very high demand given how quickly the business environment is changing and incentivizing companies to adapt or perish. Companies around the world, big or small, are in need of experts who can help them make data-driven decisions.

Growth Potential: The average starting salary of a Data Scientist in India is Rs. 5.5 LPA. Moreover, the top 10% earn more than Rs. 10 LPA.

Why Relevel Courses?

540+ Hours of Live Learning



24 Live Business Cases to be solved

Learn from Industry Experts

Placement Mentoring

Regular Assessments and Tests

Merit-based Scholarships

Learn:
The Relevel Way

Doubt Solving Sessions

Weekly educator-led doubt solving sessions along with Teaching Assistant-led doubt solving sessions 4-5 times a week ensure that your understanding is crystal clear.

Quizzes & Assignments

Real learning happens only when we practice. Specially curated quizzes after each session for a life-long learning experience.

Periodic Proctored Tests

Track how far you have come along in your endeavour to learn through these fortnightly tests.

Dedicated Success Managers

To help you reach that finish line, and anything that comes enroute this journey.

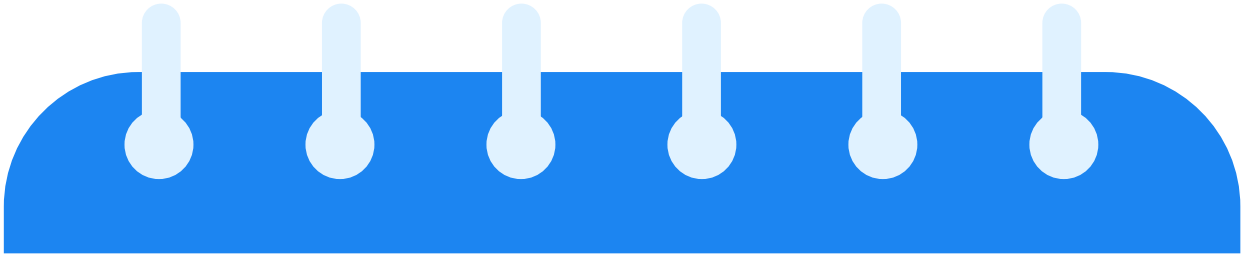
Learn by Sharing

Connect with your peers and achieve your goals together by creating a dependable community

Is this for You?

If you're looking to start your career as a Data Scientist, or to just pick up the necessary skills, you have come to the right place!

Weekly Class Schedule



MON	TUE	WED	THU	FRI	SAT	SUN
7 PM - 8 PM Doubt Clearing Session by Teaching Assistant	**Off day for practice**	**7 PM - 8 PM** Doubt Clearing Session by Teaching Assistant	**Off day for practice**	**7 PM - 8 PM** Doubt Clearing Session by Teaching Assistant	**11 AM - 2 PM*** Session by Educator	**Guest Lecture/ Event/Doubt Session with educators/ Test**
8 PM - 11 PM Session by Educator		**8 PM - 11 PM** Session by Educator		**8 PM - 11 PM** Session by Educator	**7 PM - 8 PM** Doubt Clearing Session by Teaching Assistant	

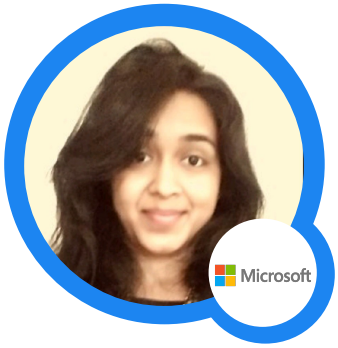
*Weekend session timings may change depending on the availability of the educator.

Classes by Experts

Rutvik Acharya
Data Scientist, Microsoft

Subhodeep Mukherjee
Data Scientist II, Amazon

Shivendra Kumar
Data Scientist, Money Tap Pvt. Ltd.



Priyal Doshi
Sr. Data Scientist, Microsoft

Garima Garg
Business Analyst, Airbnb

Learn from Industry Leaders

Kamal Mishra
Co Founder & Director, Accelerate Ai

Eshan Tiwari
Data Analytics Lead, Facebook

Curriculum Snippet

Module 1

Introduction to Data Science 3 weeks

1. **Introduction to Data science and Basic Maths**

   If there is one thing that is common among all Data Scientists, it is their ability to crunch numbers for successful decision-making. Through this module, we make sure that you learn this art early in your journey through an intensive 3 - week aptitude and data interpretation learning program.

2. **Quantitative Aptitude**

   Understand how to read datasets and draw underlying conclusions that help in analytical decision-making. In addition, improve your ability to solve logical puzzles that frequently appear in the aptitude rounds of Data Science interviews.

   - Percentages, Ratio and Proportion
   - Partnership, Average, Profit and Loss, Simple Interest and Compound Interest
   - DI - Caselets and Tables
   - DI - Bar Graphs, Column Graphs, Venn Diagrams

Module 2

Calculus and Algebra 1 week

1. Data Science is powered by four critical concepts of mathematics, which includes, statistics, probability, calculus and algebra. While statistics and probability are the core of every model, calculus and algebra helps in learning and optimization.

   - Introduction to Limits
   - Basics of Calculus - Differentiation and Integration
   - Introduction to algebra
   - Linear algebra

Module 3

Statistics and Probability — 3 weeks

1. Learn basics of Data Science including distributions, probability, discrete random variables, sampling, hypothesis testing & regression which would lay the foundation for your career as a Data Scientist and beyond.

    - Sampling, Tendency and distribution of data
    - Hypothesis Testing, Variation, Regression
    - Probability distribution, Normal distribution
    - Poisson's theory, Bayes's theory
    - Central limit, One sample, T - Test
    - ANOVA and chi - square

Module 4

Excel — 2 weeks

1. You will also gain a deep understanding of widely - used data analysis and presentation tools like Excel and Google sheets.

    - Fundamentals of Excel and basic functions
    - Dashboard preparation
    - VBA and Macros
    - Advanced Excel

Module 5

SQL — 3 weeks

1. **Intro to SQL:**

SQL is used in programming and designed for managing data held in a relational database management system. Learn to use SQL to extract and analyze data stored in databases amongst other skills like Filtering, Sorting & Join.

- Filtering, Sorting, and Extracting data via SQL
- Subqueries and Joins

2. **Advanced SQL**

- Different functions in SQL
- Indexes, Stored Procedures & Stored Objects
- Database Triggers
- SQL Capstone Project

Module 6

Python — 4 weeks

1. **Learn Basics of Python for Data Science**

Start learning Python from the very basic level. Develop an understanding of how Python helps in exploratory data analysis, which includes data profiling, visualizing results, and creating observations to shape the next steps in the analysis. Also, learn how to work with data and automate processes that will make you an efficient Data Scientist.

- Python Data Structures
- Python Programming Fundamentals

2. **Learn how to work with Data in Python**

Nobody likes doing routine repeatable tasks. This is where Python comes to help because it allows us to automate many processes. Learn how to work with data and automate processes that will make your life easier.

- Working with Data in Python
- Importing Data in Python

Module 6 (contd.)

3 Learn Exploratory Data Analysis and Model Development

- Data Wrangling in Python - Exploratory Data Analysis in Python
- Model Development in Python

4 Python Modules for Data Analysis

- Python Basics
- Webscrapping
- Python custom functions
- Lambda Function - Regular Expressions
- Data science Life cycle - NumpyModule
- Data science Life cycle - PandsModule

Module 7

Exploratory Data Analysis 2 weeks

1. In this module, learn the critical process of performing initial investigations on data to discover patterns, spot anomalies, test hypotheses, and check assumptions with the help of summary statistics & libraries like Pandas, etc.

- Visualisation & Data Cleaning
- Imputation Techniques
- Data analysis using Pandas
- Data analysis using Seaborn
- Data analysis using matplotlib

Module 8

Advanced Python and its implementation — 2 weeks

1 Master Data Visualization in Python

After mastering the basics, move to the next level of specialised visualisations followed by a project in Python.

- Introduction to Data Visualization Tools in Python
- Basic and Specialized Visualization Tools in Python
- Advanced Visualizations and Geospatial Data in Python

2 Apply Python in a Project

Learn all that have you learned in a real-life industry project.

- Project in Python - Neighborhood Segmentation and Clustering
- Project in Python - The Battle of Neighborhoods

Module 9

Exploratory Data Analysis — 3 weeks

1 Tableau

Understand the basics of Tableau in this module. It is a powerful and fast-growing data visualization tool used in the Business Intelligence Industry. Tableau helps create data that can be understood by professionals at any level in an organisation. It also allows non - technical users to create customised dashboards.

2 Master Data Visualization in Power BI

Power BI offers data preparation and discovery, interactive dashboards, and rich visualizations in one solution, and its self-service capabilities make it an intuitive tool for interacting with data and turning it into insights more easily.

- Prepare data for analysis and Model data in Power BI
- Model and Visualize data in Power BI
- Data analysis in Power BI and Manage workspaces and datasets in Power BI

Module 10

Data Science using PySpark — 1 week

1. Pyspark is one of the important skills to be gathered by a Data Scientist as not only it allows you to introspect on the subject matter but also saves time and is efficient. It provides an API to work with large scale datasets in a large scale computing environment.

    - Introduction to Big Data and Apache Spark
    - Apache Spark framework and RDDs
    - PySpark SQL and Data Frames
    - Introduction to Hive

Module 11

Machine Learning — 8 weeks

1. Data Scientists are supposed to have a thorough understanding of machine learning for quality predictions and analysis. This module will explain to you the importance of Machine Learning and why every Data Scientist must need it.

    - Introduction to Machine Learning, Applications of Machine Learning, Supervised vs Unsupervised Learning, Stages of ML Project

2. Supervised learning

    - CART, KNN (classifier, distance metrics, KNN regression)
    - Decision Trees (hyper parameter, depth, number of leaves), Naive Bayes

3. Unsupervised Learning

    - Clustering - K-Means & Hierarchical, Distance methods - Euclidean, Manhattan, Cosine, Mahalanobis, Features of a Cluster - Labels, Centroids, Inertia
    - Eigen vectors and Eigen values, Principal component analysis

Module 11 (contd.)

4. Enabling Techniques
   - Bagging & Boosting
   - AdaBoost & Gradient boosting
   - Random Forest
5. Time Series
   - Classification of Time Series Techniques (Univariate & Multivariate), Time Series Modeling & Forecasting Techniques, Seasonal Decomposition
   - ARIMA/ARIMAX/SARIMA/SARIMAX, Regression , Evaluation of Forecasting Models
   - Linear and logistic regression
   - Deployment (Rshiny and Streamlit)

Module 12

Cloud Computing and NLP — 1 week

1. **Cloud Computing**

   Cloud computing is becoming increasingly vital for Data Scientists. Cloud computing makes expanding computing power and deploying data solutions much easier and is therefore handy for Data Scientists who are digging into large datasets.

   - What is Cloud Computing? Why does it matter? Traditional IT Infrastructure vs. Cloud Infrastructure
   - Cloud Companies (Microsoft Azure, GCP, AWS) & their Cloud Services (Compute, storage, networking, apps, cognitive etc.)
   - Overview of Cloud Segments: IaaS, PaaS, SaaS, Overview of Cloud Deployment Models
   - Overview of Cloud Security, AWS vs. Azure vs. GCP, Implementation of ML/DL model in Cloud

2. **NLP**

   Sentiment extraction, Lexicon and Emotion mining

Module 13

Text Mining 1 week

1. Text mining is important as it helps in fetching the information from an unstructured or semi-structured text that is stored in natural language using advanced analytics and statistical algorithms by the Data Scientist.

    - Text cleaning, Regular expressions, Stemming, Lemmatization
    - Word cloud, Principal Component Analysis, Bigrams & Trigrams
    - Web scraping, Text summarization, Lex Rank algorithm
    - Latent Dirichlet Allocation (LDA) Technique, Word2vec Architecture (Skip Grams vs CBOW)
    - Text classification, Document vectors, Text classification using Doc2vec

Module 14

Capstone Projects and Business communication 1 week

1. For the final module, get prepared for your interviews by solving various real-life case studies, business problems and guesstimate type questions. Develop a charismatic personality and learn how to effectively communicate in a professional setting through multiple personality development sessions.

Capstone Projects

Use your learnings to solve real business problems faced by some major companies.

Build a chatbot

Chatbots play an important role in any business. Use techniques backed by Python & ML to handle a barrage of customer queries & messages without a slowdown.

meesho

Build a Recommendation System

Take several metrics into consideration, such as age, previously watched shows, genre, frequency, etc, and feed them into a Machine Learning model which recommends movies to the customers.

NETFLIX

Prediction Model

Use data sets like financial situation, customer behavior, spending habits, etc. and develop a ML model to predict future shopping needs of a customer.

Flipkart

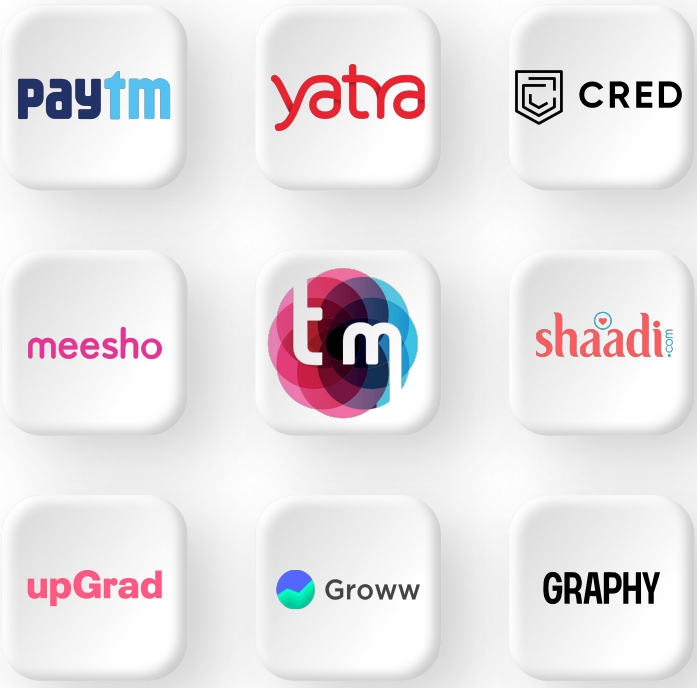
Placement and Career Support

With Relevel, you can now get a job in three simple steps:

Qualify the Relevel Data Analytics Test

Interview with a minimum of 5 companies

Receive an offer within 15 days!



How to Apply?

Here is all you need to do to kickstart your dream career.